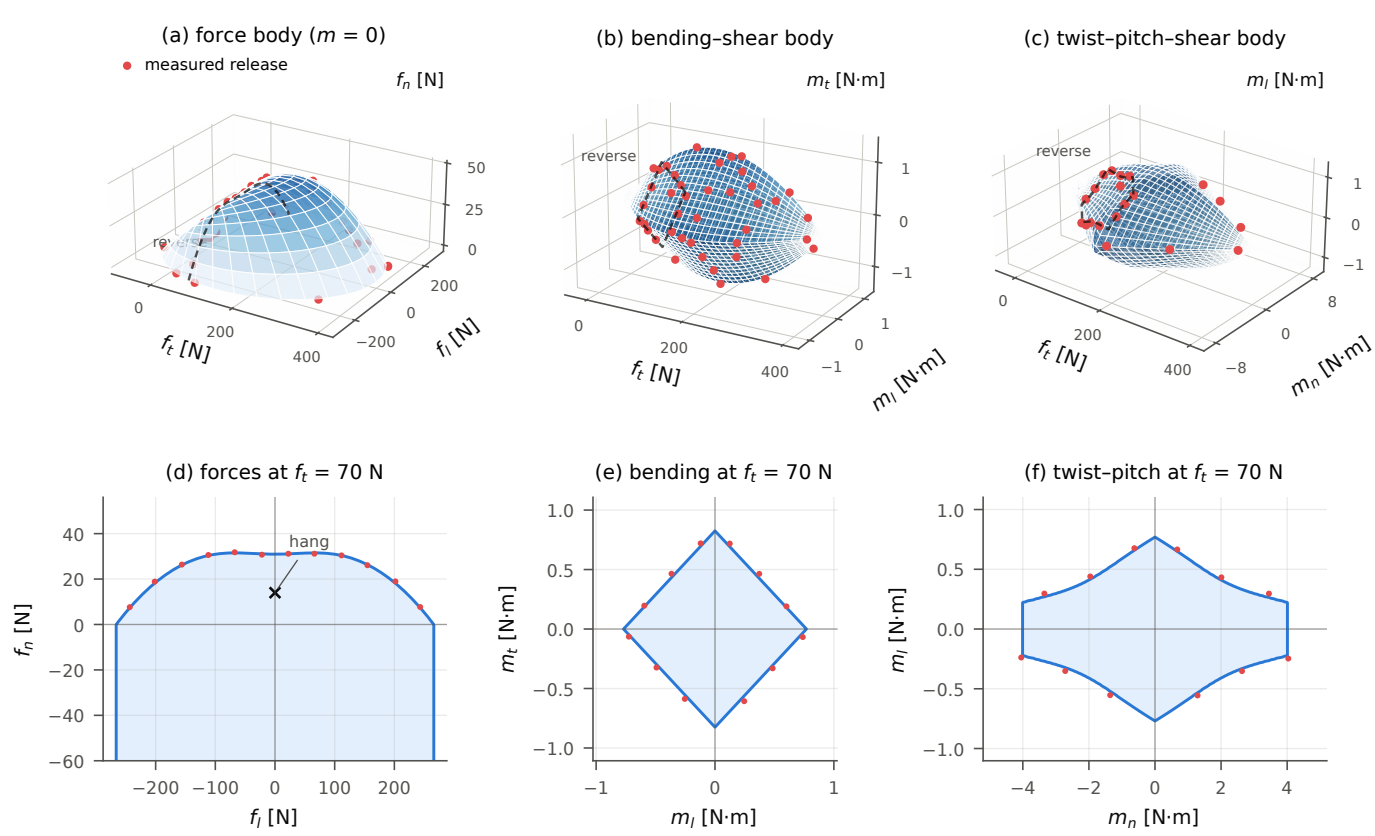

(a) force body (m = 0)
measured release
f_n [N]
reverse
50
25
0
200
0
−200
f_l [N]
0
200
400
f_t [N]
(b) bending–shear body
m_t [N·m]
reverse
1
0
−1
1
0
−1
m_l [N·m]
0
200
400
f_t [N]
(c) twist–pitch–shear body
m_l [N·m]
reverse
1
0
−1
8
0
−8
m_n [N·m]
0
200
400
f_t [N]
(d) forces at f_t = 70 N
hang
40
20
0
−20
−40
−60
f_n [N]
−200
−100
0
100
200
f_l [N]
(e) bending at f_t = 70 N
1.0
0.5
0.0
−0.5
−1.0
m_t [N·m]
−1
0
1
m_l [N·m]
(f) twist–pitch at f_t = 70 N
1.0
0.5
0.0
−0.5
−1.0
m_l [N·m]
−4
−2
0
2
4
m_n [N·m]